Changes Since Previous Submissions

Compression Ratio Is Not a Serving Metric: A Measurement Study of KV-Cache Compression for Long-Context LLM Serving

Previous submissions. This work was submitted twice to *Transactions on Machine Learning Research* (TMLR) during 2026. The first submission was titled after the evaluation harness itself; we do not reproduce that title here, because the harness is public under that name and quoting it would deanonymise this submission. The second was titled "Beyond Compression Ratio: Evaluating the Algorithmic and Systems Utility of Long-Context KV-Cache Compression". The submission identifiers are available to the chairs on request. Both were desk-rejected by the editors, and in neither case was the paper sent out for review: **no external reviewer reports were produced and no technical feedback of any kind was received.** The notices stated that the submission was judged unlikely to meet one or both of TMLR's criteria and could not be advanced given submission volume and limited reviewer and action-editor capacity.

We state this plainly because it determines how this document should be read: **none of the changes below responds to a review**, because there was none. They are our own reassessment of how the work should be framed and for which community. Both earlier versions were written as descriptions of an evaluation framework; we now think the framework was the least interesting thing about the results, and that the work belongs at a systems venue as a measurement study.

What changed.

- **Re-centred from a benchmark to a systems finding**, with a new title, abstract and contribution ordering. The paper leads with a measured claim about serving — that compression ratio and task score are not sufficient to predict prefill latency, decode throughput and resident memory — and introduces the methodology only after that claim is established; both earlier versions introduced the framework first. The contributions now lead with the empirical characterisation rather than the composite scores.
- **The first two pages were rebuilt around that finding**, in recognition of the rapid-review round: by the end of page 2 the reader has the systems problem, why conventional reporting is insufficient for it, the three quantitative findings, and their consequences for capacity planning and latency objectives; the explicit contribution bullets follow immediately on page 3.
- **A conceptual problem in the metric exposition was fixed.** The earlier versions argued that representation utility and serving efficiency are distinct axes while reporting a composite $\mathcal{S}_{\text{sys}}$ that contains $\mathcal{S}_{\text{res}}$ by construction. The pure hardware term ϕ is now the serving axis, $\mathcal{S}_{\text{sys}}$ is stated to be a presentational convenience and never evidence, and the reported rank displacements are identified as a lower bound. No formula and no measured value changed.
- **Statistical language was scoped to the evidence**: claims of the form "none predicts the others" became statements about what is and is not detectable at this sample size, while direct matched observations are stated as the direct observations they are.
- **Related work was rewritten for a systems audience**, and now engages concurrent work that reaches a compatible headline conclusion from a different measurement design, stating what our design adds to it.
- **Length was reduced to 11 pages by content editing**, not by spacing or font changes: redundant exposition was cut and secondary material (proofs, alternative formulations, per-task and per-context detail, protocol, artifact appendix) moved to appendices.
- **Figures were regenerated at final printed size** so every label renders at 8 pt or larger without scaling, and captions now state the systems takeaway rather than describe the axes.
- **An artifact appendix was added**, and the manuscript was anonymised for double-blind review: the harness is referred to by a pseudonym because it is public under its own name, and prior work by the authors is cited in the third person.

What did not change. No new experiments were run and no measured value was altered. The study is the same 3,150 paired query executions over the same seven methods, three models, five tasks and context lengths to 128K. Every number, table and figure is regenerated from the same released per-query records, and a verification script re-derives each quantity quoted in the prose and fails if the manuscript has drifted from the data. The limitations — single GPU, batch size one, 2B–7B models, and the implementation caveats for the two methods whose production kernels were unavailable on our platform — are unchanged and remain stated in the body.